Alright Madhura, you're asking for the **full lowdown on every command and concept**, step by step, so you can really *understand* what's going on and not just memorize. I'll break it down **category by category**, and explain each command and why it's used, with examples.

1⃣ Linux Fundamentals

Linux OS Basics

- **What Linux is:** An open-source operating system (OS) based on UNIX.
- **Why used in servers:** Stable, secure, lightweight, flexible, free. Most cloud servers run Linux.
- **Linux distributions:** Variants of Linux with different package managers and default tools:
  - **Ubuntu/Debian:** Popular for ease of use.
  - **CentOS/RHEL:** Enterprise servers, stable and secure.
  - **Differences:** Package management (apt vs yum), default config files, systemctl vs service commands.
- **Kernel vs User Space:**
  - **Kernel:** Core of OS, interacts directly with hardware.
  - **User Space:** Where applications and commands run.

File System Structure

Directory	Purpose
/	Root directory, top of the file system
/home	User home directories
/etc	System configuration files
/var	Variable data like logs, databases
/tmp	Temporary files
/usr	User programs, libraries, docs
/bin	Essential binary executables

Directory	Purpose
/sbin	System admin binaries (root-only)

- **File types:**
  - **Regular file:** normal text or data file.
  - **Directory:** folder.
  - **Symbolic link:** shortcut to another file.
  - **Device files:** represent hardware.
  - **Sockets:** for inter-process communication.
- **Hidden files:** Start with .; e.g., .bashrc.

File & Directory Management Commands

Command	What it does	Example
ls	List files in a directory	ls -l (detailed), ls -a (includes hidden files)
cd	Change directory	cd /etc
pwd	Print working directory	Shows full path
mkdir	Make a new directory	mkdir project_test
rmdir	Remove an empty directory	rmdir old_folder
rm	Remove files or folders	rm file.txt, rm -r folder (recursive)
cp	Copy files or directories	cp file.txt file_backup.txt, cp -r folder1 folder2
mv	Move or rename	mv file.txt notes.txt
tree	Show folder structure as tree	tree /home/user

- **Absolute vs relative path:**
  - **Absolute:** full path from root /home/user/file.txt
  - **Relative:** relative to current directory ../file.txt

File Permissions & Ownership

- Permissions: **r = read, w = write, x = execute**
- Viewing permissions: ls -l
- -rw-r--r-- 1 user group 123 Jan 1 file.txt
  - First char: file type (- regular, d directory)
  - Next 3: owner permissions
  - Next 3: group permissions
  - Last 3: others
- Commands:
  - chmod 644 file.txt → change permissions
  - chown user file.txt → change owner
  - chgrp group file.txt → change group
- **Special permissions:**
  - **Sticky bit:** only owner can delete file in directory (chmod +t)
  - **setuid:** file runs as owner (chmod u+s file)
  - **setgid:** file/group inherits parent group (chmod g+s folder)

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2 Common Linux Commands

Viewing & Editing Files

Command	Use
cat file.txt	Show full file content
more file.txt	Paginate file (scroll with space)
less file.txt	Like more, but scroll up/down
head file.txt	Show first 10 lines
tail file.txt	Show last 10 lines
nano file.txt	Open file in nano editor (easy)
vi file.txt	Open file in vi editor (advanced)

Searching & Finding

Command	Use
grep "text" file.txt	Search for text inside file
find /path -name "*.txt"	Find files by name
locate file.txt	Quickly locate files (database must be updated)
which command	Path of executable (e.g., which python)
whereis command	Shows location of binary, source, and man page

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Process Management

Command	Use
ps	Show current processes
top	Interactive view of CPU, memory usage
htop	Advanced top (optional)
kill PID	Stop a process
killall process_name	Kill all processes with that name
jobs	List background jobs
bg	Resume job in background
fg	Bring background job to foreground

Disk & System Info

Command	Use
df -h	Disk usage of filesystems
du -sh folder	Size of folder
free -m	Memory usage in MB
uptime	System uptime and load
uname -a	System info (kernel, OS)

Networking Basics

Command	Use
ping 8.8.8.8	Check connectivity
ifconfig or ip addr	Show IP addresses and interfaces
netstat -tulnp or ss -tulnp	Show listening ports

Compression & Archiving

Command	Use
tar -cvf file.tar folder/	Archive folder
tar -xvf file.tar	Extract archive
gzip file	Compress file
gunzip file.gz	Decompress file
zip file.zip folder/	Compress into zip
unzip file.zip	Extract zip

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3 User & Group Management

Command	Use
whoami	Show current user
id	Show UID, GID, groups
who	Show logged-in users
useradd trainee1	Add new user
usermod -aG group user	Add user to group
passwd user	Change password
groupadd group1	Create group
groups user	Show user’s groups

- **sudo:** Run commands as root.
- **root:** Superuser, all privileges.

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4 Package Management

- **Ubuntu/Debian:**
  - sudo apt update → update package list
  - sudo apt upgrade → upgrade installed packages
  - sudo apt install package → install package
  - sudo apt remove package → uninstall package
- **RHEL/CentOS:**
  - sudo yum install package → install
  - sudo yum update → upgrade

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5 Bash Scripting Basics

- **Shebang:** #!/bin/bash → tells OS to use bash
- **Variables:** NAME=Madhura
- **Loops:**

```
for i in {1..5}; do
  echo $i
done
```

- **Conditionals:**

```
if [ -f file.txt ]; then
  echo "File exists"
fi
```

- **Read user input:**

```
read NAME
echo "Hello $NAME"
```

- **Run script:** bash script.sh or ./script.sh

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6 Logs & Monitoring

- cat /var/log/syslog → View system logs
- tail -f /var/log/messages → Monitor logs in real-time

- top → See CPU/memory usage
- df -h → Check disk space
- du -sh /folder → Folder size
- free -m → Memory usage
- Identify high-CPU processes: top or ps -aux --sort=-%cpu | head

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7 Version Control (Bonus)

- git clone <repo> → Copy repo
- git add file → Stage changes
- git commit -m "message" → Commit changes
- git push → Push to remote

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8 Soft Skills

- Explain commands and steps clearly.
- Be curious, ask questions.
- Take notes of what you did — documentation is valued.

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9 Hands-on Practice Checklist

- Create/move/delete folders/files
- Change permissions/ownership
- Navigate /etc, /var, /home
- Create simple bash scripts
- Install/remove a package
- Check disk, memory, CPU, processes
- Monitor logs
- Add/change users and groups

Madhura, if you want, I can **make a full table of "Task → Command → Expected Output → Notes"** for **all these commands**,

so you can literally **practice everything like a lab sheet**. It'll make you interview-ready in **1 week**.

Do you want me to make that for you?